Three Problems on the Mostar Index and Albertson Irregularity*

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Abstract

This paper revisits three open problems at the interface of distance-unbalancedness and degree irregularity. The first concerns the least Mostar index among trees with a prescribed degree sequence. Starting from the Deng–Li reduction to V-shaped caterpillars, we give an exact dynamic program which computes the optimum and reconstructs all extremal trees. The second concerns the largest possible gap between the Mostar index and Albertson's irregularity on trees of a fixed order. We express the problem through centroidal transmission and degree irregularity, and obtain a finite rooted-branch recursion determining the exact value and all extremal trees for every order. The third concerns elementary edge operations. For arbitrary connected graphs we derive exact formulae for the change of the gap under edge addition and under deletion of a non-cut edge, and we exhibit small examples showing that no monotonicity principle holds in general. Implementation details, comparison with known small-order data, and special-class formulae are included for reproducibility.

Keywords. Mostar index; Albertson irregularity; prescribed degree sequence; extremal trees; edge addition and deletion.

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1 Introduction

Topological indices are among the standard numerical descriptors in chemical graph theory. A molecular graph records atoms as vertices and chemical bonds as edges, and a bond-additive descriptor assigns to a molecule a sum of edge contributions. Distance-based descriptors are particularly sensitive to the way vertices are distributed around a bond. The Mostar index, introduced by Došlić, Martinjak, Škrekovski, Tipurić Spužević, and Zubac [8], is such a descriptor: it measures, edge by edge, how far the two sides of a graph are from being metrically balanced. This interpretation is natural both for molecular graphs and for general graphs.

All graphs considered in this paper are finite, simple, and connected. For an edge $uv \in E(G)$, let

$$N_u(uv) = \{x \in V(G) : d_G(x,u) < d_G(x,v)\}, \qquad N_v(uv) = \{x \in V(G) : d_G(x,v) < d_G(x,u)\},$$

and put $n_u(uv) = |N_u(uv)|$ and $n_v(uv) = |N_v(uv)|$. The Mostar index is

$$\mathrm{Mo}(G) = \sum_{uv \in E(G)} |n_u(uv) - n_v(uv)|.$$

Thus the summand attached to uv compares the numbers of vertices lying closer to the two endpoints of the bond uv. Equivalently, since adjacent vertices have distances differing by at most one, the same summand is the absolute difference between the transmissions of u and v.

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Hence $\mathrm{Mo}(G)$ may also be regarded as the total edge variation of the transmission function. It is zero exactly for distance-balanced graphs.

Albertson's irregularity is the corresponding degree-based edge sum

$$\mathrm{irr}(G) = \sum_{uv \in E(G)} |d_G(u) - d_G(v)|,$$

introduced in [1]. The two indices have parallel forms, but they measure different phenomena. The Mostar index is global and metric; a single edge contribution depends on distance comparisons involving the whole graph. Albertson's irregularity is local; its contribution depends only on the degrees of the endpoints. The difference

$$\Delta_M(G) = \mathrm{Mo}(G) - \mathrm{irr}(G)$$

therefore measures the excess of metric edge imbalance over local degree imbalance. Gao, Xu, and Došlić [9] initiated the study of this difference. They proved, in particular, that $\mathrm{Mo}(T) \geq \mathrm{irr}(T)$ for every tree T, with equality precisely for the star, and they proposed several open problems on extremal values and elementary graph operations. The present paper addresses three problems in this circle.

The first problem is the minimization of the Mostar index among trees with a prescribed degree sequence

$$DS = (d_1, d_2, \ldots, d_p, 1, 1, \ldots, 1), \qquad d_1 \geq d_2 \geq \cdots \geq d_p \geq 2.$$

This is a natural fixed-valence problem for acyclic molecular graphs. Deng and Li [6] solved the corresponding maximum problem and proved that every minimum tree must be a caterpillar whose non-pendent degrees appear along the spine in a V-shaped order. They also settled the cases with three and four non-pendent vertices. Starting from their structural reduction, Theorem 4.1 completes the remaining finite optimization by an exact dynamic program over admissible V-caterpillars. The same recursion, with backtracking, recovers all minimizing trees.

The second problem concerns the maximum of $\Delta_M(T)$ over all trees of a fixed order. Gao, Xu, and Došlić [9] asked for a characterization of the extremal n-vertex trees and supplied exhaustive data for $7 \leq n \leq 22$. Geneson and Tsai [10] later obtained the asymptotic order of the maximum. Here we give an exact solution for every n. The key reduction shows that, for trees, maximizing $\mathrm{Mo}(T) - \mathrm{irr}(T)$ is equivalent to minimizing $2\rho(T) + \mathrm{irr}(T)$, where $\rho(T)$ is the minimum transmission of T. This turns the problem into a centroidal optimization and leads to the rooted-branch recursion of Theorem 5.2, which determines both the exact maximum and all extremal trees.

The third problem concerns one-edge operations. For Albertson's irregularity, adding or deleting an edge changes only the degrees of the affected endpoints and the contributions on edges incident with them. For the Mostar index, the same operation may change distance comparisons for many old edges. This contrast explains why Δ_M has no general monotonicity under elementary edge operations. In Section 6 we derive exact formulae for the change of Δ_M under the addition of one edge and under the deletion of one non-cut edge. We also give small examples showing that the change may be positive, negative, or zero.

We state the scope of the paper explicitly. The maximum problem for $\mathrm{Mo} - \mathrm{irr}$ is solved here in the class of trees; the corresponding extremal problem over all connected graphs, also raised in [9], remains open. Likewise, our deletion formula is restricted to non-cut edges so that connectedness is preserved. This restriction is natural in the present setting, since all distance comparisons are taken in connected graphs.

2 Background and positioning

The Mostar index has developed into an active topic in mathematical chemistry and graph theory. The survey of Ali and Došlić [2] summarizes early bounds, extremal results, variants, and

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open questions. Among the foundational papers, Alizadeh, Xu, and Klavžar studied trees and product graphs, including the effect of edge contraction [3], while Dehgardi and Azari related the Mostar index to Albertson's irregularity in graphs of small diameter [4]. Related measures of distance-unbalancedness and minimum imbalance have been considered by Miklavič and Šparl [17] and by Kramer and Rautenbach [16].

Extremal questions for the Mostar index are especially well represented in the literature. For trees, known results involve restrictions such as fixed order, maximum degree, diameter, number of pendent vertices, degree sequence, and segment sequence [12, 6, 5, 7, 14]. In particular, Deng and Li's theorem for fixed degree sequences is the structural input for our first main theorem. For graphs with cycles, extremal results are known for several families, including cacti and bicyclic or tricyclic graphs [11, 22, 13]. Recent work has also treated general bounds and extremal behaviour in bipartite graphs, split graphs, and graphs of bounded maximum degree [18, 19, 15]. These papers concern the Mostar index itself; the second and third parts of the present paper concern the comparison between the Mostar index and degree irregularity.

The invariant $\Delta_M = \mathrm{Mo} - \mathrm{irr}$ was introduced in [9]. Gao, Xu, and Došlić determined the minimum and second minimum of Δ_M over trees, obtained parity information for cacti, and studied subdivision and contraction of tree edges. Their paper also contains the finite data and the open questions that motivate the tree maximum problem and the one-edge-operation problem treated here. Geneson and Tsai's work on peripherality gives the asymptotic size of the tree maximum [10], and Tang later studied related extremal bounds for peripherality measures [21]. The rooted recursion in Theorem 5.2 strengthens the tree part of this line of work from asymptotic or bounded-order information to an exact finite characterization valid for every order.

One-edge deletion has also been studied for the Mostar index in a different setting. Shenoy, Kumar, and Suhas computed the effect of deleting a vertex or an edge on the Mostar index for several special graph families, including generalized wheels, wheels, cycles, stars, complete graphs, and complete bipartite graphs [20]. Section 6 is not a repetition of those results. The invariant here is $\mathrm{Mo} - \mathrm{irr}$, not Mo alone; the formulae are valid for arbitrary connected graphs; and edge deletion is restricted to non-cut edges in order to preserve connectedness. Their special-class computations are nevertheless useful comparison points for the Mostar component of our variation formulae.

In summary, the paper makes three contributions. It gives an exact algorithmic characterization of the minimum Mostar index for a prescribed tree degree sequence; it solves the tree maximum problem for $\mathrm{Mo} - \mathrm{irr}$ for every order; and it derives general edge-addition and non-cut-edge-deletion formulae for $\mathrm{Mo} - \mathrm{irr}$. The first two results are constructive: the associated dynamic programs compute the extremal values and recover all corresponding extremal trees.

3 Preliminaries

For a tree T of order n and an edge e, deleting e gives two components of orders s_e and $n - s_e$. Hence the Mostar contribution of e is

$$\varphi_T(e) = |n - 2s_e|. \tag{1}$$

Equivalently, if

$$\mu_T(e) = \min\{s_e, n - s_e\},$$

then

$$\mathrm{Mo}(T) = \sum_{e \in E(T)} (n - 2\mu_T(e)) = n(n-1) - 2 \sum_{e \in E(T)} \mu_T(e). \tag{2}$$

We shall also use the following theorem of Deng and Li.

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Theorem 3.1 (Deng–Li [6]). *Let*

$$DS = (d_1, d_2, \dots, d_p, 1, 1, \dots, 1), \qquad d_1 \geq d_2 \geq \cdots \geq d_p \geq 2,$$

be the degree sequence of a tree. If T has minimum Mostar index among all trees with degree sequence DS, then T is a caterpillar. Moreover, if the degrees of the non-pendent vertices along its spine are

$$(d_{i_1}, d_{i_2}, \dots, d_{i_p}),$$

then for some $r \in \{1, 2, \dots, p-1\}$,

$$d_{i_1} \geq d_{i_2} \geq \cdots \geq d_{i_r}, \qquad d_{i_{r+1}} \leq d_{i_{r+2}} \leq \cdots \leq d_{i_p}.$$

For $p = 3$ and $p = 4$ the corresponding minimizers have spine degree sequences

$$(d_1, d_3, d_2), \qquad (d_1, d_4, d_3, d_2),$$

respectively.

A caterpillar satisfying the monotonicity condition in Theorem 3.1 will be called an admissible V-caterpillar.

4 Trees with a fixed degree sequence

Let

$$DS = (d_1, d_2, \dots, d_p, 1, 1, \dots, 1), \qquad d_1 \geq d_2 \geq \cdots \geq d_p \geq 2,$$

and set

$$c_i = d_i - 1 \quad (1 \leq i \leq p), \qquad C = \sum_{i=1}^{p} c_i = n - 2.$$

The number of leaves is $n - p$. In any caterpillar with degree sequence DS, each leaf edge contributes $n - 2 = C$ to the Mostar index. Therefore the total contribution of all leaf edges is the fixed number

$$(n - p)(n - 2). \tag{3}$$

It remains to minimize the contribution of the spine.

Suppose that the spine degrees are represented by an ordering

$$x_1, x_2, \dots, x_p$$

of $c_1, c_2, \dots, c_p$, where x_j is one less than the degree of the j-th non-pendent spine vertex. If

$$S_i = x_1 + x_2 + \cdots + x_i,$$

then the component on the left of the i-th spine edge has order $1 + S_i$. Since $n = C + 2$, the i-th spine edge contributes

$$|n - 2(1 + S_i)| = |C - 2S_i|.$$

Thus the spine contribution is

$$\Phi(x_1, \dots, x_p) = \sum_{i=1}^{p-1} |C - 2S_i|. \tag{4}$$

By Theorem 3.1, it is enough to minimize (4) over admissible V-orderings.

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Theorem 4.1. *For $i = 0, 1, \ldots, p$, let*

$$P_i = \sum_{j=1}^{i} c_j, \qquad P_0 = 0.$$

If $p = 1$, then the unique tree is a star and

$$\mathrm{Mo}_{\min}(DS) = (n-1)(n-2).$$

Assume now that $p \geq 2$. Define $A_i(s)$ as follows: after processing $c_1, \ldots, c_i$, the value $A_i(s)$ is the minimum uncorrected spine cost among all assignments in which the elements assigned to the left side have total sum s. Let

$$A_0(0) = 0, \qquad A_0(s) = +\infty \quad (s \neq 0).$$

For $i \geq 1$,

$$A_i(s) = \min\{A_{i-1}(s - c_i) + |C - 2s|, A_{i-1}(s) + |C - 2(P_i - s)|\}, \tag{5}$$

where infeasible terms are interpreted as $+\infty$. Then the minimum Mostar index among all trees with degree sequence DS is

$$\mathrm{Mo}_{\min}(DS) = (n-p)(n-2) + \min_{1 \leq s \leq C-1} \{A_p(s) - |C - 2s|\}. \tag{6}$$

Moreover, all minimizers are obtained from the optimal dynamic-programming paths: if an optimal path produces a partition

$$\{1, 2, \ldots, p\} = L \sqcup R,$$

where $L = \{\ell_1 < \ell_2 < \cdots < \ell_k\}$ and $R = \{r_1 < r_2 < \cdots < r_m\}$, then the spine degree sequence is

$$(d_{\ell_1}, d_{\ell_2}, \ldots, d_{\ell_k}, d_{r_m}, d_{r_{m-1}}, \ldots, d_{r_1}), \tag{7}$$

up to reversal and repetitions caused by equal degrees.

Proof. If $p = 1$, the tree is the star $K_{1,n-1}$, and every one of its $n-1$ edges contributes $n-2$ to the Mostar index.

Let $p \geq 2$. By Theorem 3.1, a minimizer is an admissible V-caterpillar. Choosing such a caterpillar is equivalent to assigning each c_i to the left side or to the right side of the spine. On the left side the indices are used in increasing order, while on the right side they appear in decreasing order in the final spine order.

Assume that after processing $c_1, \ldots, c_i$ the left-side total is s. If c_i is put on the left side, then a new left prefix of value s appears and contributes $|C - 2s|$. If c_i is put on the right side, the corresponding right-side prefix sum is $P_i - s$, and its contribution is $|C - 2(P_i - s)|$. This gives the recurrence (5).

At the end, if the total left sum is s, the left side contributes the central term $|C - 2s|$, and the right side contributes the identical central term $|C - 2(C - s)| = |C - 2s|$. In the spine these two terms correspond to the same central cut and must be counted only once. This explains the correction $-|C - 2s|$ in (6). Adding the fixed leaf contribution (3) proves the value formula.

Finally, tracing all choices attaining the minima in (5) and (6) gives exactly the partitions $L \sqcup R$ producing (7). Conversely, every admissible V-caterpillar corresponds to such a partition. Hence all minimizers are obtained. $\square$

Remark 4.2. The dynamic program has $O(pC)$ states and constant transition cost if all $A_i(s)$ are stored in a table. It is therefore pseudo-polynomial in the order of the tree and exact for every fixed degree sequence.

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5 Maximizing $\mathrm{Mo}(T) - \mathrm{irr}(T)$ over trees

Let T be an n-vertex tree and let $\rho(T)$ denote its minimum transmission:

$$\rho(T) = \min_{v \in V(T)} \sum_{x \in V(T)} d_T(v, x).$$

A vertex attaining this minimum is a median of T. In a tree, the medians are exactly the centroids, equivalently the vertices c for which every component of $T - c$ has order at most $n/2$.

Lemma 5.1. *For every tree T,*

$$\sum_{e \in E(T)} \mu_T(e) = \rho(T).$$

Proof. Let c be a centroid of T and orient every edge away from c. For an oriented edge, the component not containing c has order at most $n/2$, so its order is precisely $\mu_T(e)$. Summing these component orders over all edges counts each vertex x exactly $d_T(c, x)$ times. Therefore the sum is $\sum_x d_T(c, x) = \rho(T)$. ☐

Combining (2) with Lemma 5.1 gives

$$\Delta_M(T) = \mathrm{Mo}(T) - \mathrm{irr}(T) = n(n-1) - \big(2\rho(T) + \mathrm{irr}(T)\big). \tag{8}$$

Thus maximizing $\Delta_M(T)$ is equivalent to minimizing $2\rho(T) + \mathrm{irr}(T)$.

5.1 Rooted branch recursion

Consider a rooted branch with s vertices whose root is adjacent to an external parent vertex of degree a in the full tree. Let $P_a(s)$ be the minimum possible contribution of this branch to $2\rho + \mathrm{irr}$, including the edge to the parent. Here all degrees are degrees in the final unrooted tree.

If $s = 1$, the branch consists of a single leaf, and

$$P_a(1) = 2 + |a - 1|. \tag{9}$$

For $s \geq 2$, let b be the degree of the root in the full tree. Then the root has $b - 1$ child branches of sizes $s_1, \ldots, s_{b-1}$ with

$$s_1 + \cdots + s_{b-1} = s - 1, \qquad s_i \geq 1.$$

The parent edge contributes $2s + |a - b|$, and the child branches contribute recursively. Hence

$$P_a(s) = \min_{2 \leq b \leq s} \left[2s + |a - b| + \min_{\substack{s_1 + \cdots + s_{b-1} = s-1 \\ s_i \geq 1}} \sum_{i=1}^{b-1} P_b(s_i) \right]. \tag{10}$$

Now let the centroid have degree a, and let the components after deleting it have orders $s_1, \ldots, s_a$. Each such component has order at most $\lfloor n/2 \rfloor$, and the orders sum to $n - 1$. Define

$$Q_n = \min_{\substack{1 \leq a \leq n-1 \\ s_1 + \cdots + s_a = n-1 \\ 1 \leq s_i \leq \lfloor n/2 \rfloor}} \sum_{i=1}^{a} P_a(s_i). \tag{11}$$

Theorem 5.2. *For every $n \geq 2$,*

$$\max_{|V(T)| = n} \big(\mathrm{Mo}(T) - \mathrm{irr}(T)\big) = n(n-1) - Q_n, \tag{12}$$

where Q_n is defined by (9), (10), and (11). Moreover, all extremal trees are obtained, up to isomorphism, by taking all choices that attain the minima in these recursions.

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Proof. Let T be an n-vertex tree and choose a centroid c of degree a. The components of $T - c$ have sizes $s_1, \ldots, s_a$ and all satisfy $s_i \leq \lfloor n/2 \rfloor$. In the rooted representation with root c, each edge contribution to $2\rho(T)$ is twice the order of the subtree below that edge, and the irregularity contribution is the absolute difference of the two endpoint degrees. Therefore the contribution of the i-th branch is at least $P_a(s_i)$, and hence

$$2\rho(T) + \mathrm{irr}(T) \geq \sum_{i=1}^{a} P_a(s_i) \geq Q_n.$$

By (8), this gives

$$\Delta_M(T) \leq n(n-1) - Q_n.$$

Conversely, choose a and $s_1, \ldots, s_a$ attaining (11), and recursively choose rooted branches attaining (9) and (10). The condition $s_i \leq \lfloor n/2 \rfloor$ ensures that the constructed root is a centroid. For the resulting tree T^*, equality holds in the preceding inequalities, so

$$\Delta_M(T^*) = n(n-1) - Q_n.$$

Tracing all equality cases gives all extremal trees. □

Remark 5.3. Theorem 5.2 is a finite recursive characterization. It is not meant to assert that the extremal tree is unique or belongs to one fixed elementary family for all n.

6 Edge addition and edge deletion

This section treats the third problem. We work with edge operations only, and in the deletion case the deleted edge is assumed not to be a cut edge. Vertex deletion and bridge deletion are outside the present connected-graph framework. This should be distinguished from deletion formulae for the Mostar index itself in special graph classes; see, for instance, [20].

For an oriented edge uv of a connected graph H, define

$$\chi_H(z; u, v) = \begin{cases} 1, & d_H(z,u) < d_H(z,v), \\ -1, & d_H(z,v) < d_H(z,u), \\ 0, & d_H(z,u) = d_H(z,v), \end{cases}$$

and put

$$\theta_H(u,v) = \sum_{z \in V(H)} \chi_H(z; u, v).$$

Then $|\theta_H(u,v)|$ is exactly the Mostar contribution of the edge uv. The following formulae keep track of the global distance term and the local irregularity term separately.

6.1 Exact variation formulae

Let $xy \notin E(G)$ and let $G^+ = G + xy$. For an old edge $uv \in E(G)$, set

$$\delta^+_{uv}(z) = \chi_{G^+}(z; u, v) - \chi_G(z; u, v).$$

Also define

$$\begin{aligned} I^+_G(x,y) = {} & |d_G(x) - d_G(y)| \\ & + \sum_{xz \in E(G)} \big(|d_G(x) + 1 - d_G(z)| - |d_G(x) - d_G(z)|\big) \\ & + \sum_{yz \in E(G)} \big(|d_G(y) + 1 - d_G(z)| - |d_G(y) - d_G(z)|\big). \end{aligned}$$

Thus $I^+_G(x,y) = \mathrm{irr}(G^+) - \mathrm{irr}(G)$.

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Theorem 6.1. *If $xy \notin E(G)$ and $G^+ = G + xy$, then*

$$\begin{aligned}\Delta_M(G^+) - \Delta_M(G) = {} & |\theta_{G^+}(x,y)| \\ & + \sum_{uv\in E(G)} \left(\left| \theta_G(u,v) + \sum_{z\in V(G)} \delta^+_{uv}(z) \right| - |\theta_G(u,v)| \right) \\ & - I^+_G(x,y).\end{aligned} \tag{13}$$

Proof. The first term in (13) is the Mostar contribution of the new edge xy. For an old edge uv, the value of $\theta_G(u,v)$ changes by $\sum_z \delta^+_{uv}(z)$, which gives the displayed sum for the change in the old Mostar contributions. Finally, after adding xy, only the new edge and the old edges incident with x or y can change their irregularity contribution; this is precisely the quantity $I^+_G(x,y)$. Subtracting the irregularity change from the Mostar change proves the formula. □

Now let $e_0 = xy \in E(G)$ and assume that $G^- = G - e_0$ is connected. For $uv \in E(G^-)$, put

$$\delta^-_{uv}(z) = \chi_{G^-}(z;u,v) - \chi_G(z;u,v).$$

Define

$$\begin{aligned}I^-_G(x,y) = {} & -|d_G(x) - d_G(y)| \\ & + \sum_{xz\in E(G^-)} \left(|d_G(x) - 1 - d_G(z)| - |d_G(x) - d_G(z)|\right) \\ & + \sum_{yz\in E(G^-)} \left(|d_G(y) - 1 - d_G(z)| - |d_G(y) - d_G(z)|\right).\end{aligned}$$

Then $I^-_G(x,y) = \mathrm{irr}(G^-) - \mathrm{irr}(G)$.

Theorem 6.2. *If $e_0 = xy \in E(G)$ and $G^- = G - e_0$ is connected, then*

$$\begin{aligned}\Delta_M(G^-) - \Delta_M(G) = {} & -|\theta_G(x,y)| \\ & + \sum_{uv\in E(G^-)} \left(\left| \theta_G(u,v) + \sum_{z\in V(G)} \delta^-_{uv}(z) \right| - |\theta_G(u,v)| \right) \\ & - I^-_G(x,y).\end{aligned} \tag{14}$$

Proof. Deleting xy removes the Mostar contribution $|\theta_G(x,y)|$. For every remaining edge, the change of the signed distance imbalance is $\sum_z \delta^-_{uv}(z)$. The only irregularity contributions that can change are the deleted edge and the remaining edges incident with x or y, and their total change is $I^-_G(x,y)$. This proves (14). □

Corollary 6.3. *Let G be a connected graph with diameter at most two.*

1. *If $xy \notin E(G)$, then $\Delta_M(G + xy) = \Delta_M(G) = 0$.*
2. *If $e \in E(G)$ is not a cut edge and $G - e$ also has diameter at most two, then $\Delta_M(G - e) = \Delta_M(G) = 0$.*

In particular, deleting any edge from K_n leaves Δ_M unchanged.

Proof. For every connected graph of diameter at most two, the Mostar index equals Albertson's irregularity; see [4, 9]. Edge addition cannot increase the diameter, so the first assertion follows. The second assertion is immediate from the same equality applied to both G and $G - e$. □

The formulae above are exact for arbitrary connected graphs. They also show why a simple sign rule is unavailable: the irregularity variation is confined to the neighbourhoods of x and y, whereas a single new or deleted edge can alter distance comparisons for many old edges.

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6.2 No general monotonicity

Proposition 6.4. *Adding a single edge to a connected graph can increase, decrease, or leave unchanged the value of $\Delta_M(G) = \mathrm{Mo}(G) - \mathrm{irr}(G)$. Consequently, deleting a non-cut edge can also increase, decrease, or leave unchanged $\Delta_M(G)$.*

Proof. For P_3, adding the missing edge gives C_3, and both graphs have $\Delta_M = 0$.

For P_4, one has $\mathrm{Mo}(P_4) = 4$ and $\mathrm{irr}(P_4) = 2$, so $\Delta_M(P_4) = 2$. Adding the edge between the two leaves gives C_4, for which $\mathrm{Mo}(C_4) = \mathrm{irr}(C_4) = 0$. Thus the change is -2.

Finally, let $P_7 = v_1v_2 \cdots v_7$. A direct computation gives $\mathrm{Mo}(P_7) = 18$ and $\mathrm{irr}(P_7) = 2$, whence $\Delta_M(P_7) = 16$. After adding the chord v_1v_4, the resulting graph has $\mathrm{Mo} = 21$ and $\mathrm{irr} = 4$, and therefore $\Delta_M = 17$. In this case the change is $+1$.

The deletion statements are obtained by reversing these three edge additions. □

7 Computational supplements and closed formulae

We collect here computational forms of the two recursions, small-order data used as consistency checks, and several explicit formulae for special graph classes. These details are not needed for the proofs of the main theorems, but they make the results directly reproducible and clarify the relation with previously published computations.

7.1 Implementation of the dynamic program for fixed degree sequences

For Theorem 4.1, only two arrays of length $C + 1$ are required if one wants the optimum value only. Backpointers are stored only when all optimizing caterpillars are required. Let $A(s)$ be the array after the first $i - 1$ entries have been processed. Initially $A(0) = 0$ and $A(s) = +\infty$ for $s \neq 0$. After processing c_i, write $P_i = c_1 + \cdots + c_i$ and compute a new array B by

$$B(s + c_i) \leftarrow \min\{B(s + c_i), A(s) + |C - 2(s + c_i)|\},$$
$$B(s) \leftarrow \min\{B(s), A(s) + |C - 2(P_i - s)|\}.$$

The first update puts c_i on the left side of the V-caterpillar, while the second puts c_i on the right side. At the end one returns

$$(n - p)(n - 2) + \min_{1 \le s \le C-1} \{A(s) - |C - 2s|\},$$

with the star case $p = 1$ handled separately as in Theorem 4.1. Thus the memory consumption is $O(C)$ for the optimum value and $O(pC)$ when all optimal spine orders are reconstructed. The running time is $O(pC)$.

7.2 Implementation of the rooted-branch recursion

For Theorem 5.2, the direct recursion in (9)–(11) can be implemented by a sequence of min-plus partition tables. Fix an upper bound N for the order. For each parent degree a and branch order s, the values $P_a(s)$ are computed in increasing order of s. When $s = 1$, formula (9) applies. For $s \ge 2$, define, for each possible root degree b, the auxiliary table

$$F_b(k, S) = \min_{\substack{t_1 + \cdots + t_k = S \\ t_i \ge 1}} \sum_{i=1}^{k} P_b(t_i),$$

using only already computed values $P_b(t)$ with $t < s$. It satisfies

$$F_b(0, 0) = 0, \qquad F_b(k, S) = \min_{1 \le t \le S-k+1} \{F_b(k - 1, S - t) + P_b(t)\}.$$

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Then

$$P_a(s) = \min_{2 \le b \le s} \{2s + |a - b| + F_b(b - 1, s - 1)\}.$$

Finally, for each $n \le N$ one computes

$$Q_n = \min_{1 \le a \le n-1} \min_{\substack{s_1 + \cdots + s_a = n-1 \\ 1 \le s_i \le \lfloor n/2 \rfloor}} \sum_{i=1}^{a} P_a(s_i).$$

This last minimum is the same partition problem with the additional upper bound $s_i \le \lfloor n/2 \rfloor$. With backpointers in the tables $F_b(k, S)$, the construction of all extremal trees follows recursively from the equalities that attain the minima. A straightforward implementation is polynomial in N, and memory can be reduced by rolling the k-layers of each F_b table.

7.3 Small-order extremal data for $\max(\mathrm{Mo} - \mathrm{irr})$ over trees

Table 1 gives the first values obtained from Theorem 5.2. A profile

$$(a; s_1, s_2, \dots, s_a)$$

means that a centroid has degree a and that the components after deleting the centroid have orders $s_1, \dots, s_a$. The column "#ext" gives the number of non-isomorphic extremal trees obtained by following all equality choices in the rooted-branch recursion. For $7 \le n \le 22$, both the values and the numbers of extremal trees agree with the exhaustive data reported by Gao, Xu, and Došlić [9]. The profile column records centroid-profile types only; different rooted branch shapes may have the same profile.

Table 1: Small-order values for Theorem 5.2.

n	Q_n	$\max_T(\mathrm{Mo}(T) - \mathrm{irr}(T))$	#ext	centroid profiles attaining Q_n
2	2	0	1	$(1; 1)$
3	6	0	1	$(2; 1, 1)$
4	10	2	1	$(2; 1, 2)$
5	14	6	1	$(2; 2, 2)$
6	20	10	2	$(2; 2, 3)$; $(3; 1, 2, 2)$
7	24	18	1	$(3; 2, 2, 2)$
8	30	26	2	$(3; 2, 2, 3)$
9	36	36	5	$(3; 2, 2, 4)$; $(3; 2, 3, 3)$; $(4; 2, 2, 2, 2)$
10	42	48	9	$(3; 2, 2, 5)$; $(3; 2, 3, 4)$; $(3; 3, 3, 3)$; $(4; 2, 2, 2, 3)$
11	48	62	10	$(3; 2, 3, 5)$; $(3; 2, 4, 4)$; $(3; 3, 3, 4)$; $(4; 2, 2, 2, 4)$; $(4; 2, 2, 3, 3)$
12	54	78	13	$(3; 2, 4, 5)$; $(3; 3, 3, 5)$; $(3; 3, 4, 4)$; $(4; 2, 2, 2, 5)$; $(4; 2, 2, 3, 4)$; $(4; 2, 3, 3, 3)$
13	60	96	15	$(3; 2, 5, 5)$; $(3; 3, 4, 5)$; $(3; 4, 4, 4)$; $(4; 2, 2, 3, 5)$; $(4; 2, 2, 4, 4)$; $(4; 2, 3, 3, 4)$; $(4; 3, 3, 3, 3)$
14	66	116	13	$(3; 3, 5, 5)$; $(3; 4, 4, 5)$; $(4; 2, 2, 4, 5)$; $(4; 2, 3, 3, 5)$; $(4; 2, 3, 4, 4)$; $(4; 3, 3, 3, 4)$
15	72	138	12	$(3; 4, 5, 5)$; $(4; 2, 2, 5, 5)$; $(4; 2, 3, 4, 5)$; $(4; 2, 4, 4, 4)$; $(4; 3, 3, 3, 5)$; $(4; 3, 3, 4, 4)$
16	78	162	9	$(3; 5, 5, 5)$; $(4; 2, 3, 5, 5)$; $(4; 2, 4, 4, 5)$; $(4; 3, 3, 4, 5)$; $(4; 3, 4, 4, 4)$
17	84	188	7	$(4; 2, 4, 5, 5)$; $(4; 3, 3, 5, 5)$; $(4; 3, 4, 4, 5)$; $(4; 4, 4, 4, 4)$
18	90	216	4	$(4; 2, 5, 5, 5)$; $(4; 3, 4, 5, 5)$; $(4; 4, 4, 4, 5)$
19	96	246	3	$(4; 3, 5, 5, 5)$; $(4; 4, 4, 5, 5)$
20	102	278	1	$(4; 4, 5, 5, 5)$
21	108	312	1	$(4; 5, 5, 5, 5)$
22	116	346	11	$(4; 4, 5, 5, 7)$; $(4; 5, 5, 5, 6)$; $(5; 2, 4, 5, 5, 5)$; $(5; 3, 3, 5, 5, 5)$; $(5; 3, 4, 4, 5, 5)$; $(5; 4, 4, 4, 4, 5)$

7.4 Closed formulae for unicyclic graphs and cacti

The variation formulae in Section 6 are completely general. For cacti, the Mostar part itself has a particularly explicit form. We first state the unicyclic case.

Let G be a unicyclic graph whose unique cycle is

$$C = v_0 v_1 \cdots v_{k-1} v_0.$$

For each i, let T_i be the rooted tree attached at v_i after deleting the cycle edges, and put $w_i = |V(T_i)|$. For the cycle edge $e_i = v_i v_{i+1}$, indices being read modulo k, define

$$\sigma_{i,j} = \begin{cases} 1, & d_C(v_j, v_i) < d_C(v_j, v_{i+1}), \\ -1, & d_C(v_j, v_{i+1}) < d_C(v_j, v_i), \\ 0, & d_C(v_j, v_i) = d_C(v_j, v_{i+1}). \end{cases}$$

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Then the cycle-edge contribution is

$$\varphi_G(e_i) = \left| \sum_{j=0}^{k-1} \sigma_{i,j} w_j \right|.$$

If e is a tree edge in some T_i and s_e is the order of the component of $T_i - e$ not containing v_i, then e is a bridge of G and

$$\varphi_G(e) = |n - 2s_e|.$$

Consequently,

$$\mathrm{Mo}(G) = \sum_{i=0}^{k-1} \sum_{e \in E(T_i)} |n - 2s_e| + \sum_{i=0}^{k-1} \left| \sum_{j=0}^{k-1} \sigma_{i,j} w_j \right|. \tag{15}$$

The same idea gives a formula for every cactus. Let $\mathcal{C}(G)$ be the set of cycle blocks and let $\mathcal{B}(G)$ be the set of bridges. If $e \in \mathcal{B}(G)$, let s_e be one component order of $G - e$; the value $|n - 2s_e|$ is independent of which component order is chosen. If $C = v_0 v_1 \cdots v_{k-1} v_0$ is a cycle block, remove the edges of C and let $W_j(C)$ be the order of the component containing v_j. Then

$$\mathrm{Mo}(G) = \sum_{e \in \mathcal{B}(G)} |n - 2s_e| + \sum_{C \in \mathcal{C}(G)} \sum_{i=0}^{k(C)-1} \left| \sum_{j=0}^{k(C)-1} \sigma_{i,j}^{C} W_j(C) \right|, \tag{16}$$

where $\sigma_{i,j}^{C}$ is defined by comparing the two distances in the cycle C from v_j to the endpoints of the edge $v_i v_{i+1}$. Formula (15) is the one-cycle special case of (16).

7.5 Chemical graphs

For a chemical graph, all degrees are at most 4. Let m_{ij} denote the number of edges joining a vertex of degree i to a vertex of degree j, where $1 \le i < j \le 4$. Then the irregularity has the closed form

$$\mathrm{irr}(G) = m_{12} + 2m_{13} + 3m_{14} + m_{23} + 2m_{24} + m_{34}. \tag{17}$$

Thus for a chemical cactus, combining (16) and (17) gives the explicit expression

$$\Delta_M(G) = \sum_{e \in \mathcal{B}(G)} |n - 2s_e| + \sum_{C \in \mathcal{C}(G)} \sum_{i} \left| \sum_{j} \sigma_{i,j}^{C} W_j(C) \right| - (m_{12} + 2m_{13} + 3m_{14} + m_{23} + 2m_{24} + m_{34}).$$

In particular, for a chemical tree T this reduces to

$$\Delta_M(T) = n(n-1) - 2\rho(T) - (m_{12} + 2m_{13} + 3m_{14} + m_{23} + 2m_{24} + m_{34}),$$

by Lemma 5.1. These formulae are useful for testing conjectures in chemically restricted graph classes because all terms are computable directly from bridge sizes, cycle-block weights, and degree-pair counts.

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